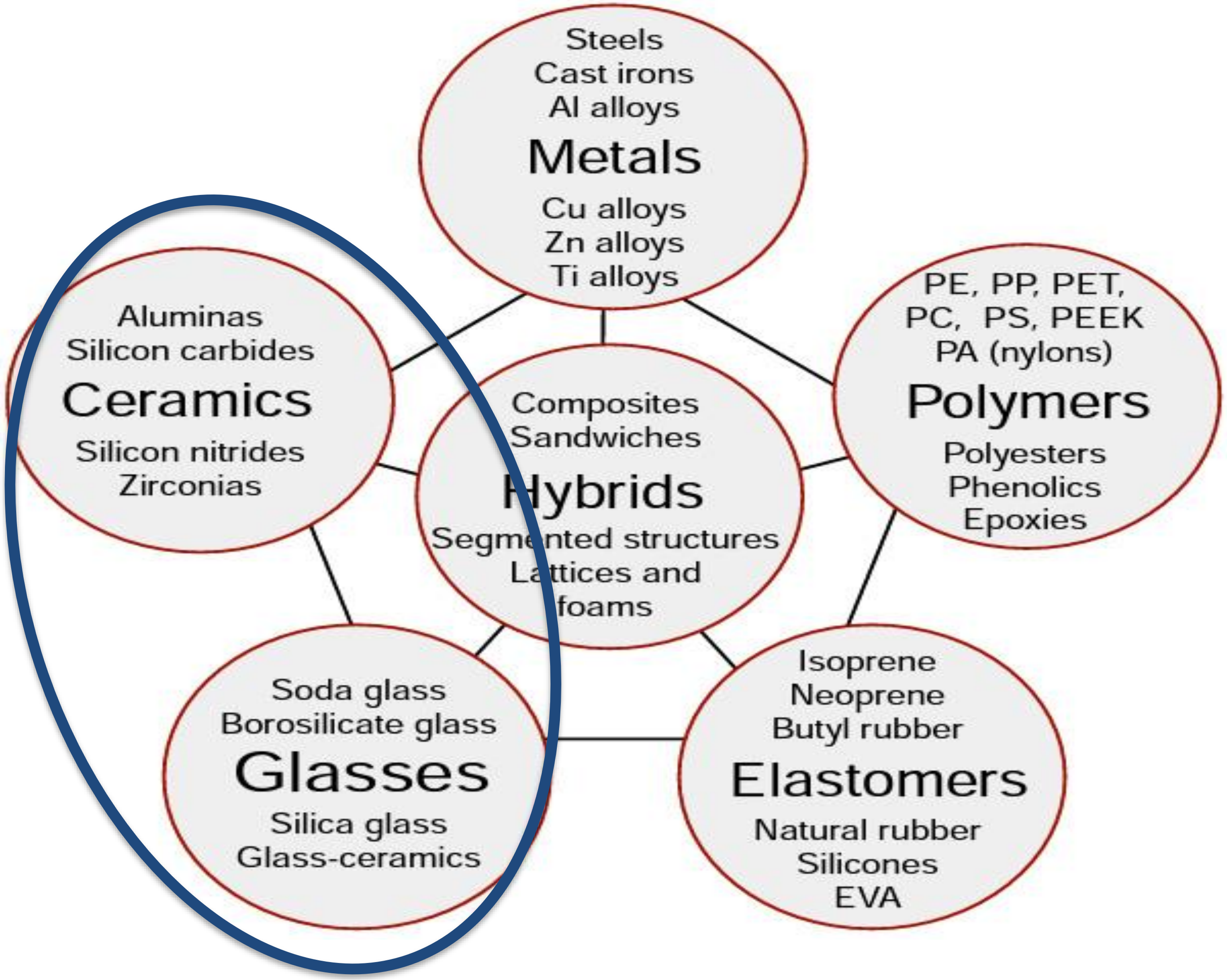
Ceramic Materials

Presented by Dr Adijat ‘Wumi Inyang

Materials Families



Materials: Engineering, Science, Processing and Design by Ashby, Shercliff and Cebon. 2007 ISBN-13: 978-0-7506-8391-3 pg 15

Week 9 lectures

In-organic – non-metallic materials

- Classification of ceramics and glasses
- Engineering ceramics
- Glass, clay and concrete
- Carbon materials

Background information:

For this week's lectures read

Callister and Rethwisch 9e Chapter 14 and Chapter 16.2
Chapter 17.8-17.11

QMPlus QXU4006 MS2 – Ceramics
QMPlus QXU4000 MS1 – Revise Crystal structures and phase diagrams

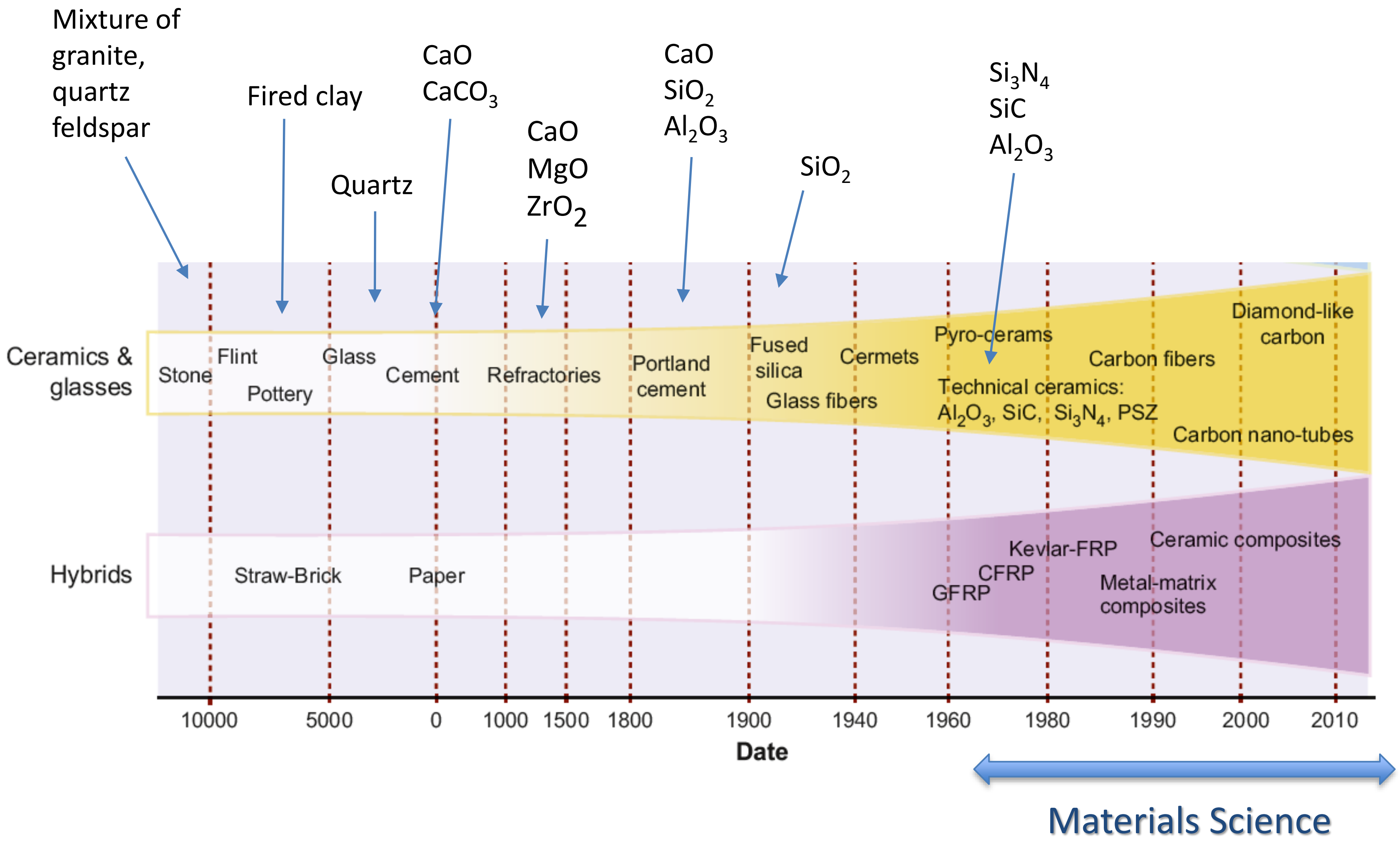
Ceramic materials

In-organic – non-metallic materials

- Porcelain ceramics
- Technical ceramics
- Refractory ceramics
- Abrasive materials
- Functional ceramics
- Glass
- Cement and concrete
- Carbon materials

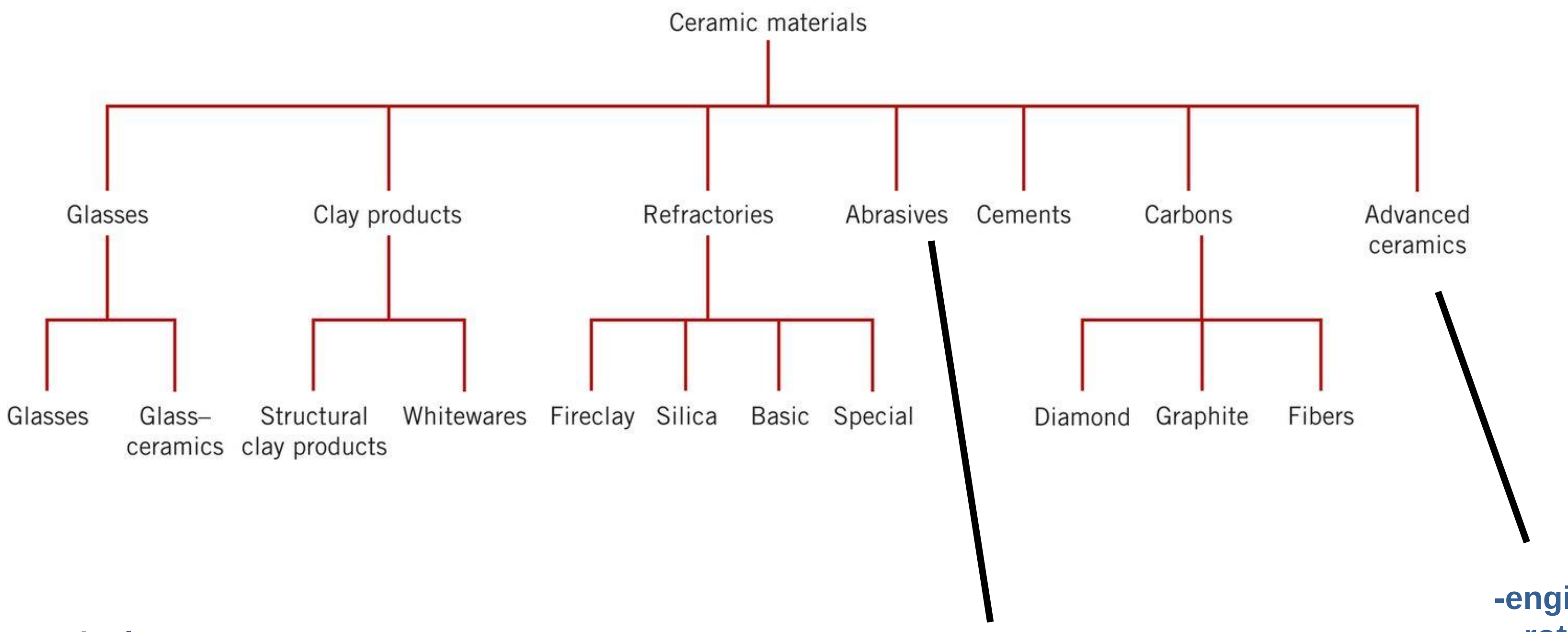
- Properties
- Processing

Evolution of ceramic components



Adapted from Ashby, 3rd Edition. Figure 1.1

Classification of ceramics



-optical
-composite reinforce
-containers/ household

-whiteware
Structural insulators

-bricks for high *T* (furnaces)

-grinding
-cutting
-polishing

-composites structural

-engine rotors valves bearings
-sensors

Applications

Adapted from Fig. 14.14 and discussion in Section 14.10-16, *Callister & Rethwisch 9e.*

Ceramic phase diagrams

MgO-Al_2O_3 system

- Magnesium aluminate **spinel** ($MgAl_2O_4$) is an excellent refractory material

  Spinel is MgO - Al_2O_3

  (= Al_2O_3 28%wt MgO)

  at high temperatures spinel is non-stoichiometric

  at lower temperatures Al_2O_3 and MgO are insoluble and separates as second phases
- There are 2 eutectic points
- Commonly used as optical **transparent ceramics** in the military industry

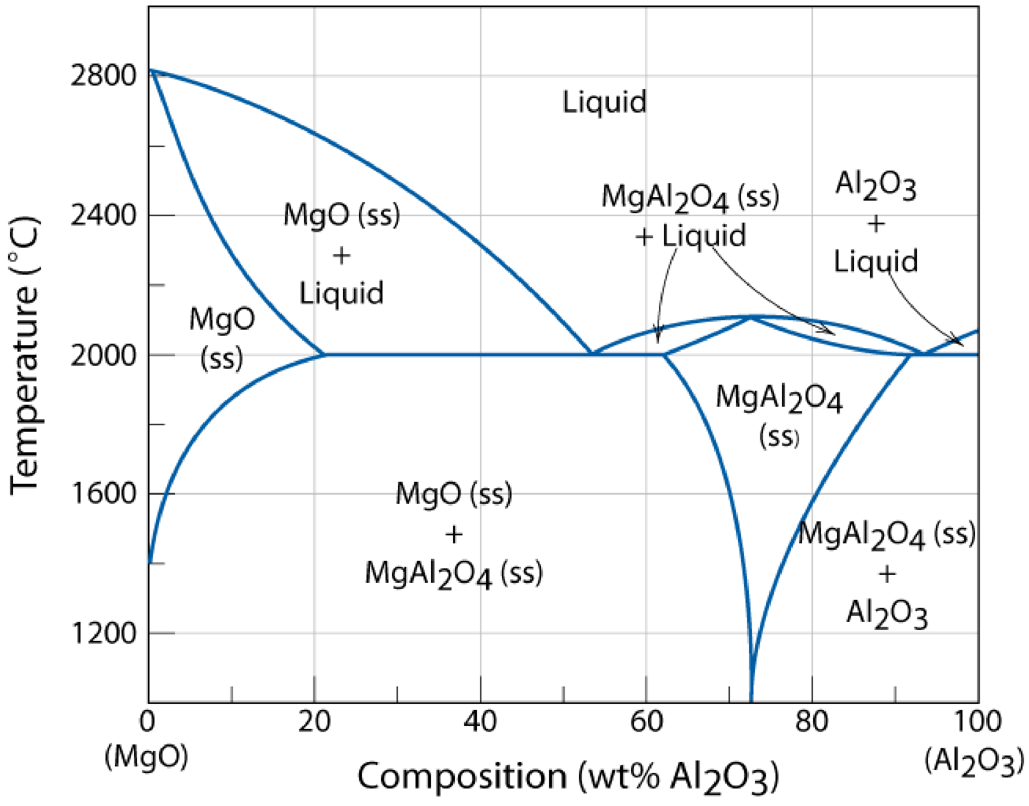


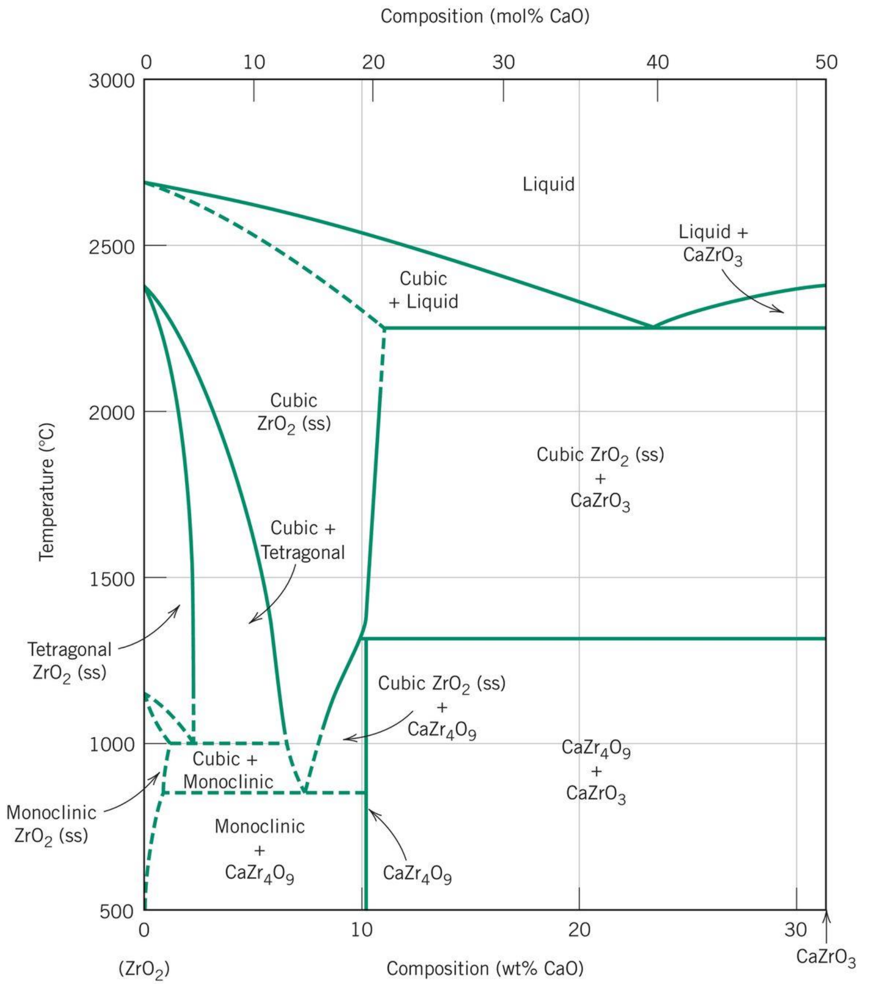
Fig. 14.2, Callister & Rethwisch 9e.
[Adapted from B. Hallstedt, "Thermodynamic Assessment of the System MgO–Al_2O_3," J. Am. Ceram. Soc., 75[6], 1502 (1992). Reprinted by permission of the American Ceramic Society.]

Ceramic phase diagrams

Zirconia-calcium system

- ZrO_2 goes through solid state transformations from cubic to tetragonal to monoclinic
- The volume change is enough to crack the material
- Adding other elements, such as CaO in 3-7%wt 'stabilizes' the cubic and tetragonal phases to room temperature by heat treatment '**Partially Stabilized Zirconia**' **(PSZ)** giving **high toughness** (4-12 MPa√m)

- Y_2O_3 and MgO may also be used to stabilize ZrO_2 phases
- Higher contents stabilize the cubic phase = **fully stabilized zirconia**



Adapted from V. S. Stubican and S. P. Ray, "Phase Equilibria and Ordering in the System ZrO2–CaO," J. Am. Ceram. Soc., 60[11–12] 535 (1977). Reprinted by permission of the American Ceramic Society.

Fig. 14.3, Callister & Rethwisch 9e.

Ceramic phase diagrams

Silica-alumina system

- SiO_2 and Al_2O_3 are not soluble in each other
- SiO_2 forms **cristobalite**
- $Al_6Si_2O_{13}$ **mullite**
- The materials are a mixture of these phases

- Main components of many ceramic refractories
- Used as **thermal insulators**

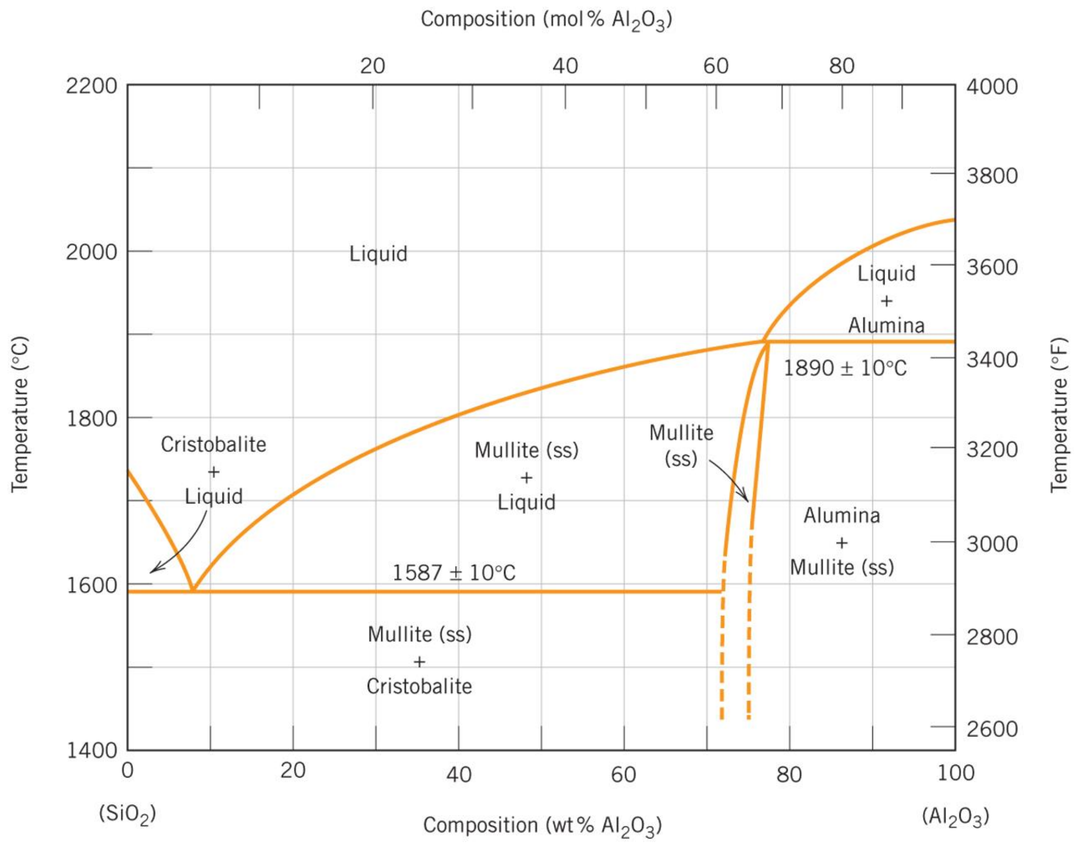


Fig. 14.4, Callister & Rethwisch 9e.

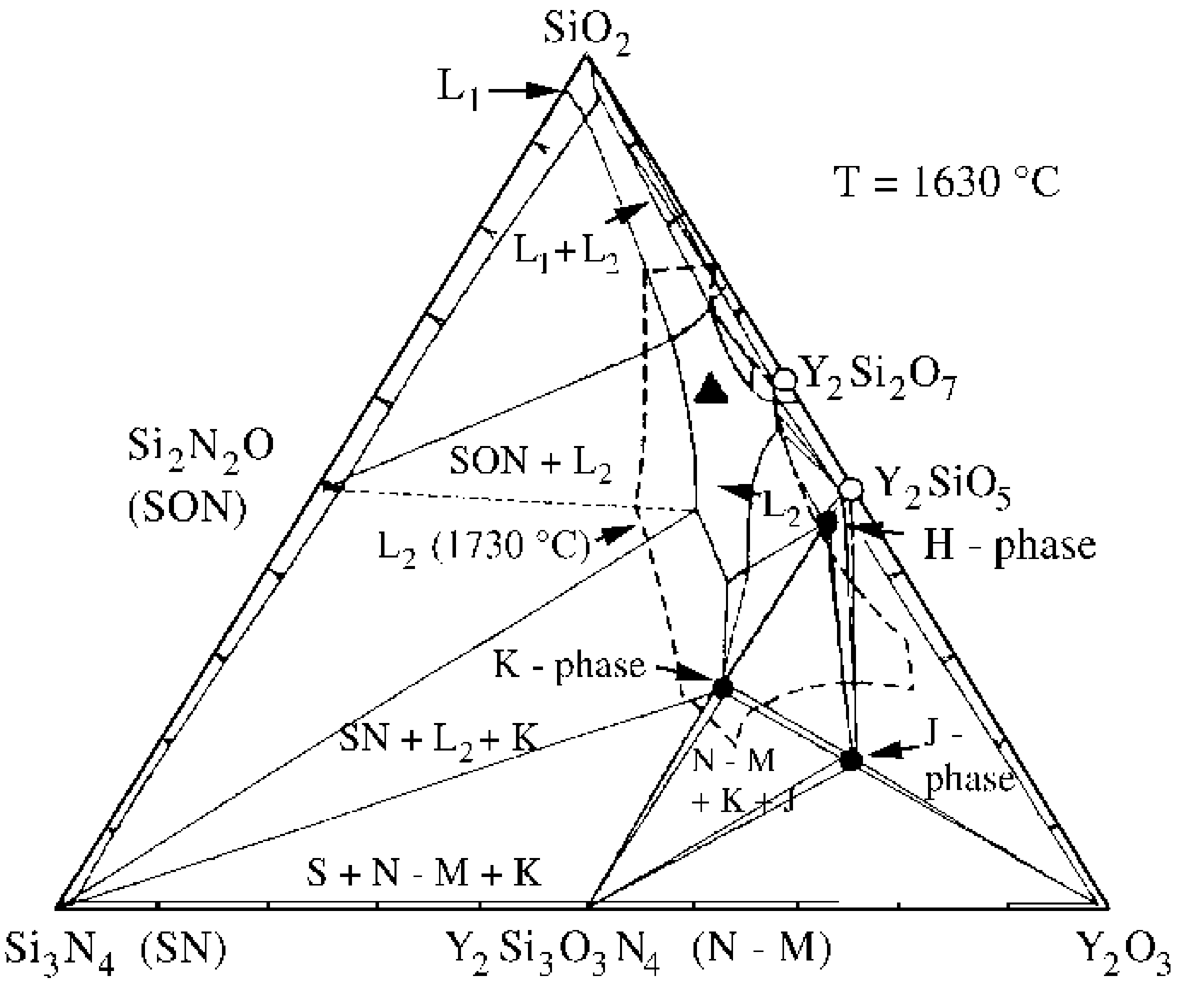
Ceramic phase diagrams

Silicon nitrides

- Many ceramic systems are tertiary (not binary)

  = 3 constituents
- The phase diagrams are **more complex**
- Elements are added to improve processing

  e.g. Y_2O_3



Ceramic phase diagrams

SiAlON ceramics

- 4 constituents
- The phase diagrams are **even more complex**

β Si_3N_4 = high toughness

YAG = solid state lasers

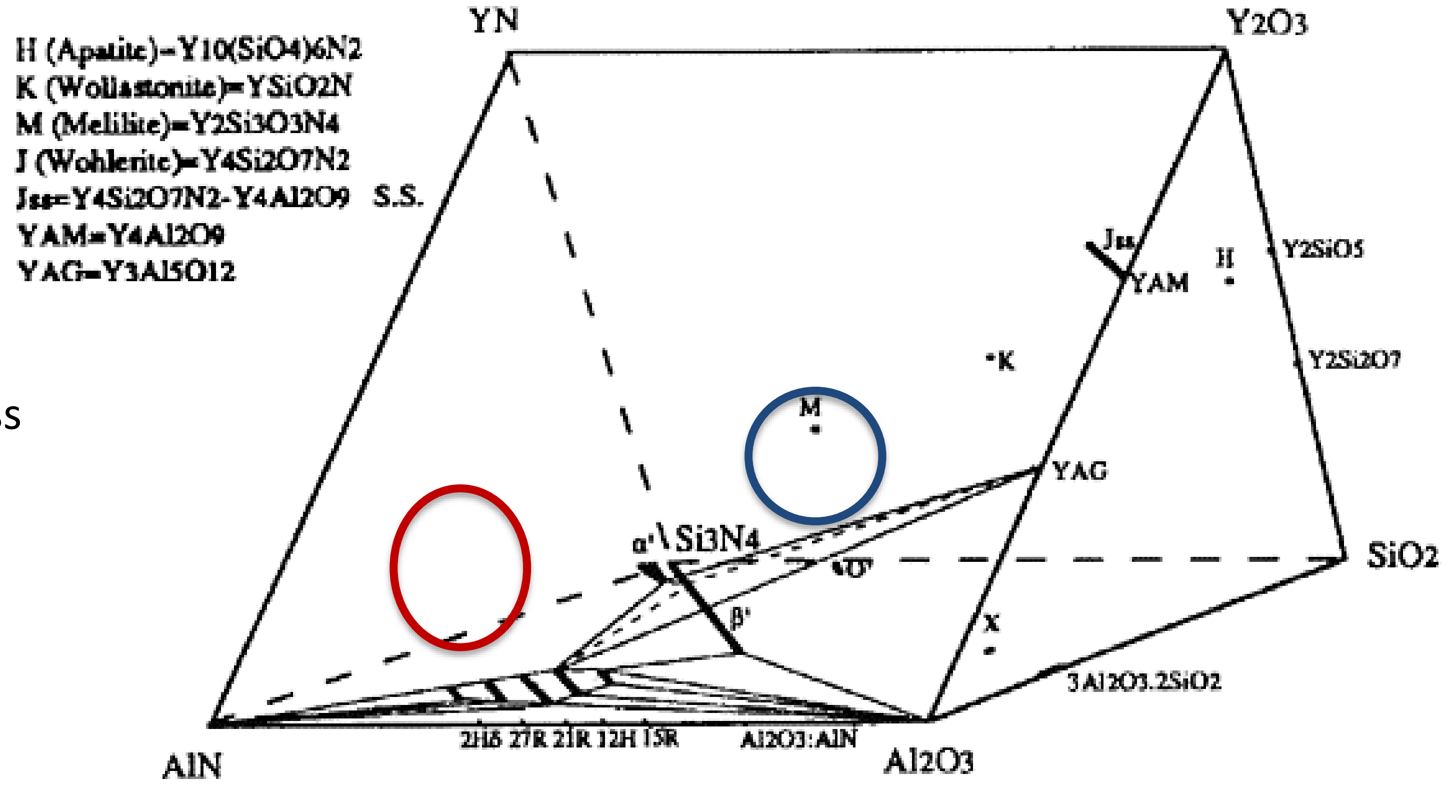
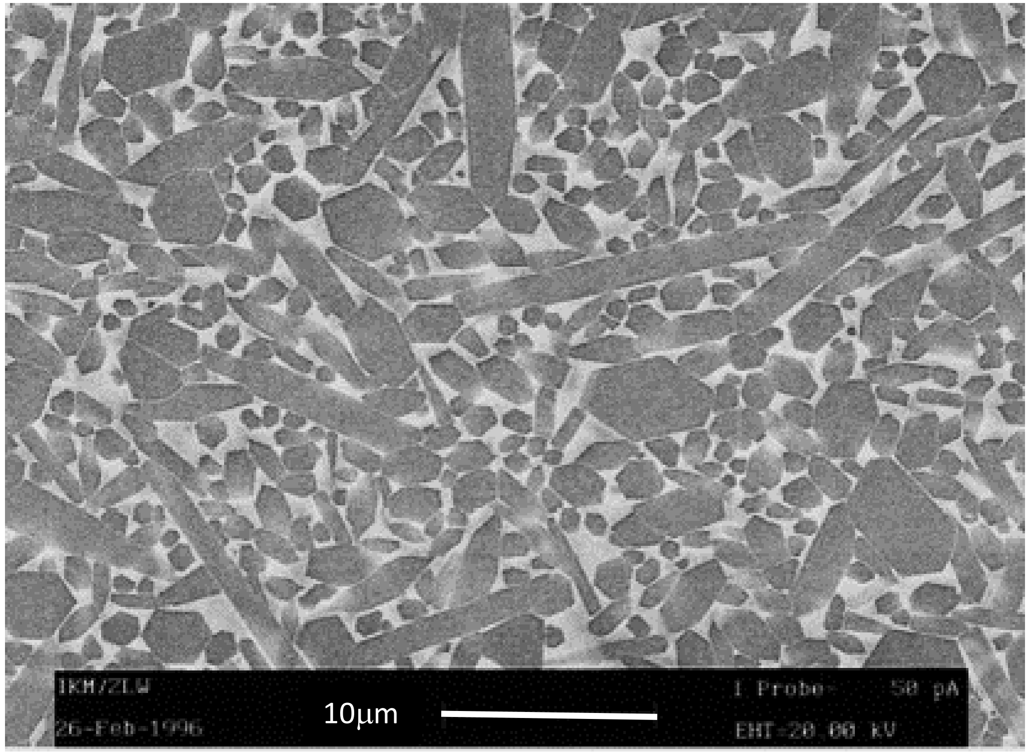


Fig 9. Izhevskiy, V. A., Gênova, L. A., & Bressiani, J. C.. (1999). Review article: RE-SiAlONa¤b- composites. *Cerâmica*, *45*(291), 05-23.

$\beta\ Si_3N_4$ beta silicon nitride

- **$\beta\ Si_3N_4$** forms needle shaped crystals = **higher toughness**
- Glassy phases form between the crystals (formed by Y_2O_3)
- Cracks follow the glassy phase = long and **tortuous crack path**



WuXi TecCeram Fine Ceramic Co., Ltd

Engineering ceramics

Different classes of ceramics:

inorganic, non-metallic materials

oxides, carbides and nitrides of Group II and III metals

- Porcelain - Kaolinite 高嶺 $Al_2Si_2O_5(OH)_4$
- Structural ceramics: Al_2O_3, SiC, Si_3N_4, SiAlON, ZrO_2
- Wear coatings: TiC, TiN
- Functional ceramics: $BaTiO_3$, $Pb[Zr_xTi_{1-x}]O_3$ (PZT), ZnO
- Abrasives: SiC, BN

Properties of ceramics

	E GPa	$\sigma_{compressive}$ MPa	Kc MPa$\sqrt{m}$	Tm K	Thermal conductivity W/m/K
Porcelain	70	350	1	~1000	1
Al_2O_3	380	3000	3-5	2323	25.6
SiC	410	2000	0.5	$\boxed{3110}$	$\boxed{84}$
Si_3N_4	310	1200	4	2173	17
ZrO_2	200	2000	$\boxed{4\text{-}12}$	2843	1.5
Diamond	$\boxed{1050}$	$\boxed{5000}$	-	-	70

Ashby and Jones II Chapter 17

Natural ceramics

Clay minerals – natural ceramics

- Kaolinite 高嶺 $Al_2Si_2O_5(OH)_4$
- Easy to form - cheap
- Thermal and electrical insulators

Porcelain ceramics

Electrical insulators

Structural ceramics

Very high temperature applications, high strength, some toughness

Aluminium Oxide (Al_2O_3) T_m = 2072°C
High temperature furnace parts
Chemically inert

Zircon Oxide (ZrO_2) T_m = 2700°C
High temperature bearings
Good toughness
Excellent surface finish

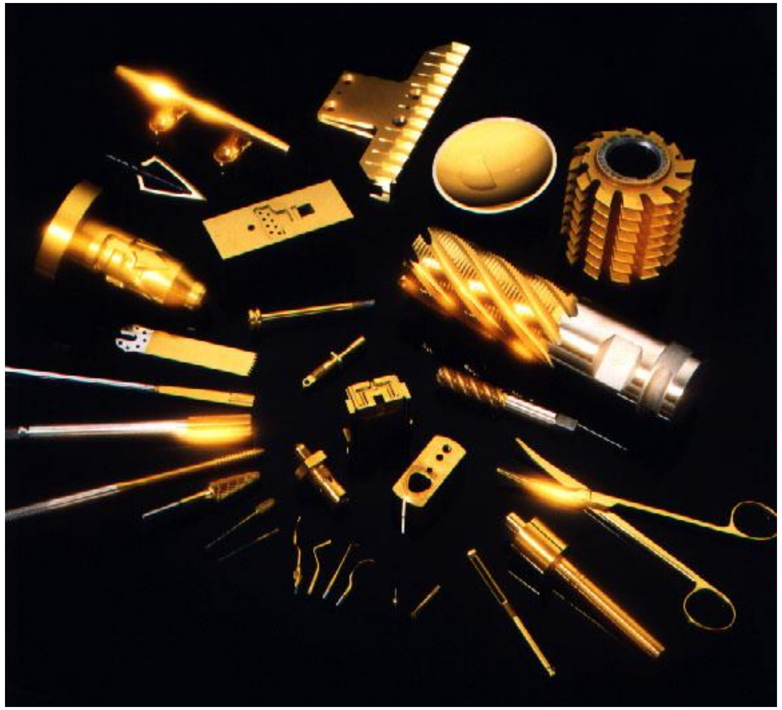
Silicon Carbide (SiC) T_m = 2830°C
Silicon Nitride (Si_3N_4) T_m = 1900°C
High temperature nozzles

Wear coatings

Thin ceramic coatings

- 1-10μm thickness,
- high hardness,
- wear resistance
- increase tool lifetime

Titanium carbide (TiC) – cutting tools

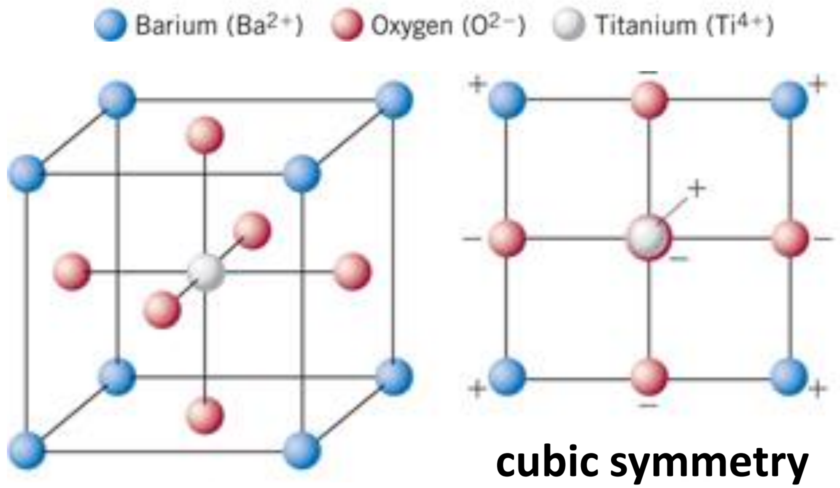


Titanium nitride (TiN) coatings

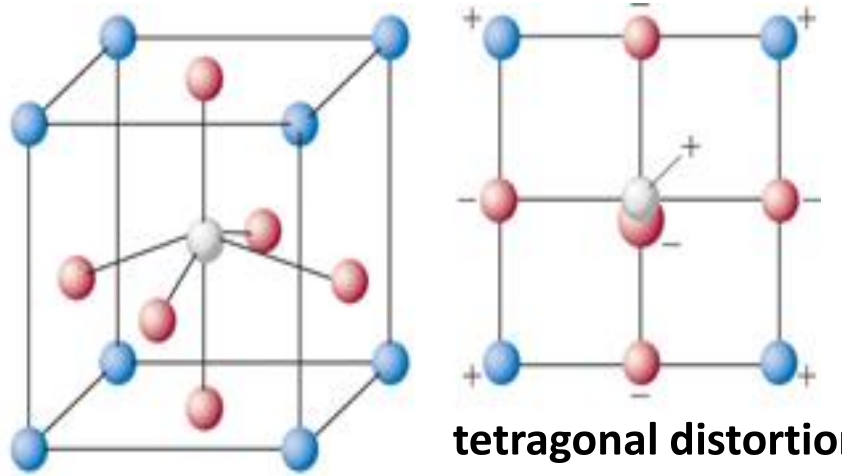
Functional ceramics

Ferro-electric materials
e.g. Barium Titanate - $BaTiO_3$

$BaTiO_3$ has the **cubic** crystal structure above 120°C (**the Curie temperature, Tc**)
Below this temperature the unit cell changes shape to **tetragonal** and the Ti^{4+} and O^{2-} ions reposition so that there is a net charge across the crystal - **polarisation**.



cubic symmetry
> 120°C



tetragonal distortion
< 120°C

Lattice constants a = b = c = 0.401nm

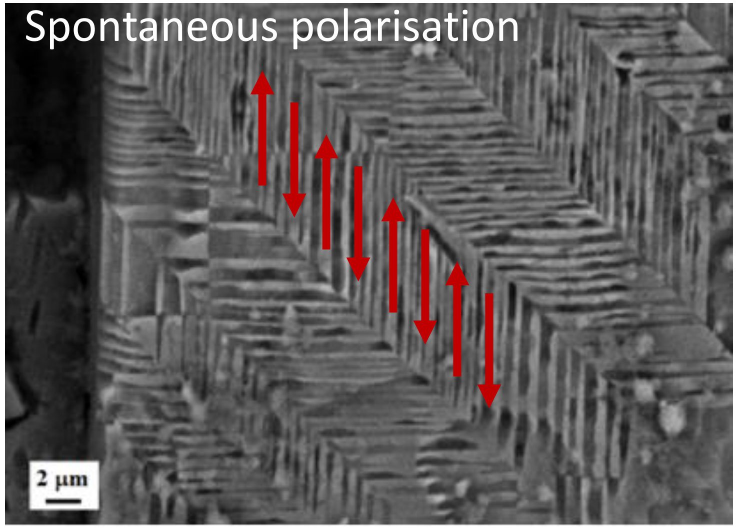
Lattice constants a = b = 0.399nm
c = 0.404nm

Functional ceramics

Ferro-electric materials
e.g. Barium Titanate - $BaTiO_3$

Ferroelectric materials **spontaneously polarize** below their Tc and form **domains** with opposite polarisation (so that the net charge is 0).

Application of an **external electric field** (**poling**) re-orients the domains so that there is a net charge across the crystal. This is used to make capacitors and sensors.



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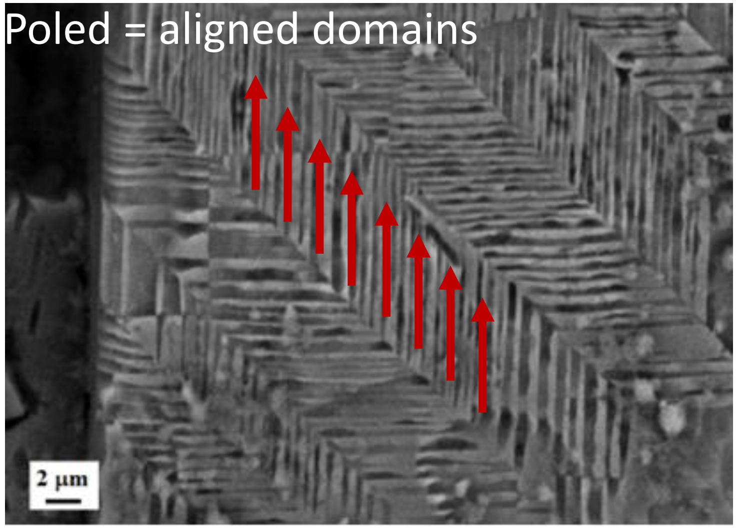

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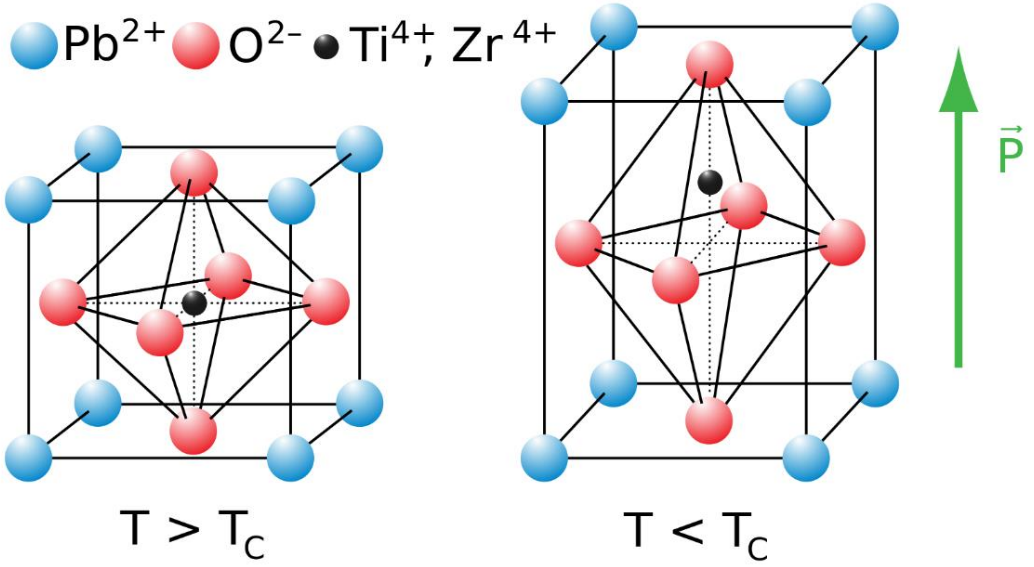
High dielectric constant, k, 2000-10,000 at 1Mhz

Functional ceramics

Piezo-electric materials
e.g. Lead zirconate titanate (PZT) – $Pb[Zr_x\ Ti_{1-x}]O_3$

Piezoelectric materials generate a **voltage** in response to a **mechanical strain**

The converse piezoelectric effect – they generate **strain** in response to an **applied voltage**



4 segment tube piezo
can bend or extend in any direction
by activating different segments

Used to make sensors and actuators

Abrasive ceramics

Grinding and material removal

high hardness

Silicon Carbide (SiC)

Density = 3160 kg/m^3

T_m = 2730°C

Hv = 24.5 GPa

Cubic boron nitride (cBN)

Density = 2100 kg/m^3

T_m = 2973°C

Hv = 46 GPa

Summary

Engineering ceramics:

- Complex crystal structures
- Complex phase behaviour
- Stabilized phases with good engineering properties
- High temperatures and high hardness